

NexGen NeuroVision Project Report

Lead Engineer: Saad Ullah

Executive Summary

This document evaluates the operational performance differences between **spatial** and **sequential neural networks** applied to raw medical imagery. We evaluated three distinct models using a comprehensive dataset of brain magnetic resonance imaging (MRI) scans. The empirical data establishes that **spatial convolutional networks fundamentally outperform sequential models for pixel-based classification tasks**.

Dataset & Preprocessing

We sourced **7,200 perfectly balanced MRI scans** across four distinct categories: **glioma**, **meningioma**, **pituitary**, and **no tumor** cases. We split the data into training, validation, and testing sets using a **66:12:22 ratio** to ensure absolute isolation and prevent data leakage.

To combat overfitting, the training pipeline deployed dynamic image augmentation including random horizontal flips, rotational shifts, and color adjustments. We resized all images to **224×224 pixels** and normalized them against standard ImageNet distribution metrics. The validation and test sets remained completely unaltered to preserve ground truth integrity.

Architectural Methodology

We engineered three separate architectures to test spatial versus sequential feature extraction paradigms.

- **CNN ResNet50:** We deployed a **ResNet50** architecture. Unlike standard transfer learning, we unfroze the entire 24 million parameter backbone. Coupled with a ReduceLROnPlateau scheduler, this allowed the network to learn the specific micro textures of brain tissue rather than relying on generic ImageNet geometric filters.
- **RNN Sequential:** We built a sequential model that flattens the image into sequences of **224 time steps**. The network processes these rows sequentially to extract temporal dependencies across the image array.
- **LSTM Sequential:** We upgraded the standard recurrent network by replacing basic cells with memory gating mechanisms, combating gradient loss over the long **224-step sequence length**.

Performance Metrics

The empirical telemetry highlights a massive gap in performance advantage for spatial models over sequential architectures across all primary key performance indicators and fine tuned CNN achieving clinical grade reliability.

Model	Accuracy	Precision	Recall	F1 Score	Parameters	Training Time (s)
CNN ResNet50	0.955	0.958099	0.955	0.953878	24,033,604	2115.86
RNN Sequential	0.65375	0.626587	0.65375	0.620363	144,196	798.27
LSTM Sequential	0.771875	0.766717	0.771875	0.759526	551,236	841.67

Visual Analytics

Area Under the Curve (AUC) Analysis

The macro-average Receiver Operating Characteristic (ROC) curves confirm overall classification reliability. The fine tuned CNN ResNet50 architecture maintains a near perfect curve while dominating both sequential architectures in false positive rejection.

Error Topography & Confusion Matrices

Confusion matrix evaluation highlights clear operational failure modes in sequential processing. **CNN ResNet50** accurately identifies the vast majority of no tumor and pituitary cases, showing minimal confusion between glioma and meningioma. Conversely, **LSTM Sequential** struggles to isolate meningioma, while **RNN Sequential** fails significantly across spatial boundaries.

Learning Progression & Hyperparameter Optimization

The training loss and validation accuracy curves validate the decision to fine tune the backbone over a 25-epoch horizon. The **ReduceLROnPlateau** scheduler successfully navigated the gradient descent, allowing the validation accuracy to stabilize near **99%** during training without triggering catastrophic forgetting or severe overfitting on the holdout set.

Final Conclusions

The empirical data proves that the fine tuned **CNN ResNet50 is the superior architecture for raw medical image classification**. While the RNN and LSTM models successfully map the image to a sequential tensor, they destroy the localized spatial geometry required to define tumor boundaries. By unfreezing the ResNet50 backbone and carefully scheduling the learning rate, the convolutional architecture successfully adapted to the complex textures of magnetic resonance imaging, achieving a highly reliable 95.5% holdout accuracy.